

# Homework: The “Deep Dive” Blueprint: Weekly Publishing Pace Dashboard

## 1. Assignment Overview

In this assignment, the manual process of creating a weekly publishing pace dashboard is analyzed to identify friction points, such as inconsistent filtering, missing URLs, incorrect publish dates, and manual chart creation. This report is pulled on a weekly basis, but it summarizes publishing progress for the current month-to-date.

The goal of the report is to show how much content has been published so far in the month, how publishing activity is distributed across brands, and which topic clusters are contributing to the current month’s output.

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## 2. Use Case: Weekly Publishing Pace Report for the Current Month

### **Problem Description:**

The current reporting process is manual, repetitive, and prone to errors because publishing data must be reviewed across multiple Google Sheets each week. The report is used to show the current month’s publishing pace and where the team stands so far. Rows must be checked for valid article URLs, “Published” status, and publish dates within the current month before the data can be summarized into brand scorecards and topic cluster visualizations.

### **Scale:**

Moderate to high volume of article rows across multiple Google Sheets or tabs.

### **Frequency:**

Weekly, with each report showing month-to-date publishing progress.

### **Manual Effort:**

Repetitive filtering, copying, grouping, and chart-building across Google Sheets data.

### **Final Output:**

A weekly dashboard showing month-to-date publishing progress, including:

- scorecards per brand
- total articles published so far this month

- article share by brand
- topic cluster breakdown by brand
- overall topic cluster breakdown
- publishing pace by week

### 3. The “As-Is” Checklist: Manual Steps

- Open the source Google Sheets that track publishing activity.
- Identify the tabs and columns needed for the report: `article_url`, `topic_cluster`, `publish_date`, `status`, and brand/source.
- Filter or review rows where `status` is “Published.”
- Check whether each published row has a valid `article_url`.
- Check whether the `publish_date` falls within the current month.
- Copy qualifying rows into a combined month-to-date dataset.
- Group the data by brand, topic cluster, and publish week.
- Manually create scorecards and charts showing publishing pace for the month.

### 4. Required Visuals: The “As-Is” Evidence

#### Visual 1: Source Google Sheet Screenshot or Mockup

A raw Google Sheet tracker showing article URL, topic cluster, publish date, status, and brand/source.

| A | B                                                                   | C                                                                                         | D                                     | E                         | F             | G                                                                 | H                                                                   | I         | J            |
|---|---------------------------------------------------------------------|-------------------------------------------------------------------------------------------|---------------------------------------|---------------------------|---------------|-------------------------------------------------------------------|---------------------------------------------------------------------|-----------|--------------|
|   | article_url                                                         | Prismic URL                                                                               | Target Keyword (slug)                 | topic_cluster             | Content Type  | Koray Content Brief                                               | Article Google Doc                                                  | status    | publish_date |
|   | <a href="https://www.xometry.com/re">https://www.xometry.com/re</a> | <a href="https://xometry-marketing.prismic.io/">https://xometry-marketing.prismic.io/</a> | basics of plastic injection molding   | injection molding         | Existing Node | <a href="https://docs.google.com/s">https://docs.google.com/s</a> | <a href="https://docs.google.com/do">https://docs.google.com/do</a> | Published | 2025-12-08   |
|   | <a href="https://www.xometry.com/re">https://www.xometry.com/re</a> | <a href="https://xometry-marketing.prismic.io/">https://xometry-marketing.prismic.io/</a> | types of injection molding technology | injection molding         | Existing Node | <a href="https://docs.google.com/s">https://docs.google.com/s</a> | <a href="https://docs.google.com/do">https://docs.google.com/do</a> | Published | 2025-12-08   |
|   | <a href="https://www.xometry.com/re">https://www.xometry.com/re</a> | <a href="https://xometry-marketing.prismic.io/">https://xometry-marketing.prismic.io/</a> | injection mold components             | injection molding         | Existing Node | <a href="https://docs.google.com/s">https://docs.google.com/s</a> | <a href="https://docs.google.com/do">https://docs.google.com/do</a> | Published | 2025-12-08   |
|   | <a href="https://www.xometry.com/re">https://www.xometry.com/re</a> | <a href="https://xometry-marketing.prismic.io/">https://xometry-marketing.prismic.io/</a> | gas assist injection molding          | injection molding         | Existing Node | <a href="https://docs.google.com/s">https://docs.google.com/s</a> | <a href="https://docs.google.com/do">https://docs.google.com/do</a> | Published | 2025-12-08   |
|   | <a href="https://www.xometry.com/re">https://www.xometry.com/re</a> | <a href="https://xometry-marketing.prismic.io/">https://xometry-marketing.prismic.io/</a> | thin wall injection molding           | injection molding         | New Node      | <a href="https://docs.google.com/s">https://docs.google.com/s</a> | <a href="https://docs.google.com/do">https://docs.google.com/do</a> | Published | 2025-12-08   |
|   | <a href="https://www.xometry.com/re">https://www.xometry.com/re</a> | <a href="https://xometry-marketing.prismic.io/">https://xometry-marketing.prismic.io/</a> | polypropylene                         | materials                 | Existing Node | <a href="https://docs.google.com/s">https://docs.google.com/s</a> | <a href="https://docs.google.com/do">https://docs.google.com/do</a> | Published | 2025-12-08   |
|   | <a href="https://www.xometry.com/ca">https://www.xometry.com/ca</a> | <a href="https://xometry-marketing.prismic.io/">https://xometry-marketing.prismic.io/</a> | liquid silicone rubber lsr            | injection molding service | Existing Node | <a href="https://docs.google.com/s">https://docs.google.com/s</a> | <a href="https://docs.google.com/do">https://docs.google.com/do</a> | Published | 2025-12-08   |
|   | <a href="https://www.xometry.com/re">https://www.xometry.com/re</a> | <a href="https://xometry-marketing.prismic.io/">https://xometry-marketing.prismic.io/</a> | foam injection molding                | injection molding         | New Node      | <a href="https://docs.google.com/s">https://docs.google.com/s</a> | <a href="https://docs.google.com/do">https://docs.google.com/do</a> | Published | 2025-12-08   |

#### Visual 2: Manual Filtering Example

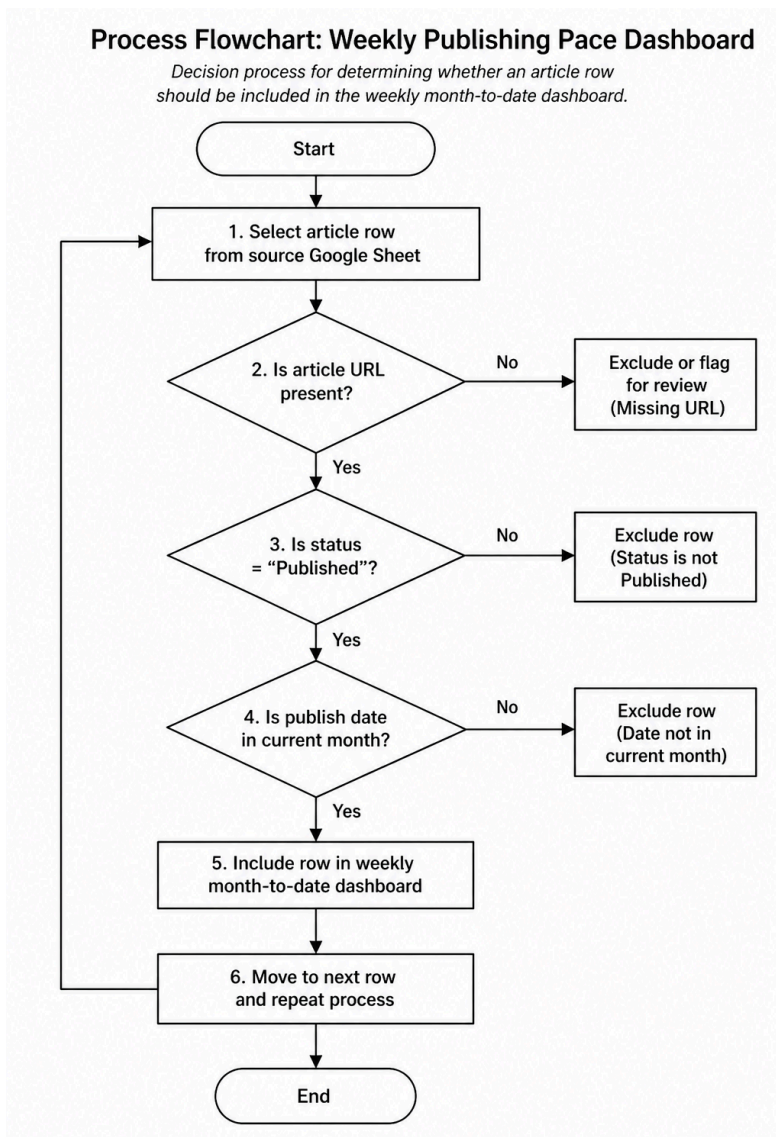
A table showing which rows are included or excluded based on URL, status, and whether the publish date falls within the current month-to-date reporting window.

| A                                                                                                                                                                       | B            | C               | D | E | F                  | G               |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------|-----------------|---|---|--------------------|-----------------|
| article_url                                                                                                                                                             | publish_date | COUNTA of artic |   |   | topic_cluster      | COUNTA of topic |
| <a href="https://www.xometry.com/resources/3d-printing/3d-printer-calibration/">https://www.xometry.com/resources/3d-printing/3d-printer-calibration/</a>               | 2026-04-01   | 1               |   |   | 3d printing        | 2               |
| <a href="https://www.xometry.com/resources/3d-printing/sketchup-free-vs-sketchup-pro/">https://www.xometry.com/resources/3d-printing/sketchup-free-vs-sketchup-pro/</a> | 2026-04-16   | 1               |   |   | machining          | 20              |
| <a href="https://www.xometry.com/resources/machining/blind-hole-in-engineering/">https://www.xometry.com/resources/machining/blind-hole-in-engineering/</a>             | 2026-04-16   | 1               |   |   | materials          | 12              |
| <a href="https://www.xometry.com/resources/machining/cnc-m-codes/">https://www.xometry.com/resources/machining/cnc-m-codes/</a>                                         | 2026-04-01   | 1               |   |   |                    |                 |
| <a href="https://www.xometry.com/resources/machining/cnc-programming/">https://www.xometry.com/resources/machining/cnc-programming/</a>                                 | 2026-04-01   | 1               |   |   |                    |                 |
| <a href="https://www.xometry.com/resources/machining/cnc-vs-manual-machining/">https://www.xometry.com/resources/machining/cnc-vs-manual-machining/</a>                 | 2026-04-01   | 1               |   |   |                    |                 |
| <a href="https://www.xometry.com/resources/machining/drilling-machines/">https://www.xometry.com/resources/machining/drilling-machines/</a>                             | 2026-04-01   | 1               |   |   |                    |                 |
| <a href="https://www.xometry.com/resources/machining/electrical-discharge-machining/">https://www.xometry.com/resources/machining/electrical-discharge-machining/</a>   | 2026-04-01   | 1               |   |   |                    |                 |
| <a href="https://www.xometry.com/resources/machining/g-code-and-m-code-differences/">https://www.xometry.com/resources/machining/g-code-and-m-code-differences/</a>     | 2026-04-16   | 1               |   |   |                    |                 |
| <a href="https://www.xometry.com/resources/machining/parts-of-a-cnc-machine/">https://www.xometry.com/resources/machining/parts-of-a-cnc-machine/</a>                   | 2026-04-01   | 1               |   |   |                    |                 |
| <a href="https://www.xometry.com/resources/machining/sinker-edm-machining/">https://www.xometry.com/resources/machining/sinker-edm-machining/</a>                       | 2026-04-01   | 1               |   |   |                    |                 |
| <a href="https://www.xometry.com/resources/machining/steel-framing/">https://www.xometry.com/resources/machining/steel-framing/</a>                                     | 2026-04-23   | 1               |   |   |                    |                 |
|                                                                                                                                                                         |              |                 |   |   | <b>Grand Total</b> | <b>34</b>       |

### Visual 3: Process Flowchart

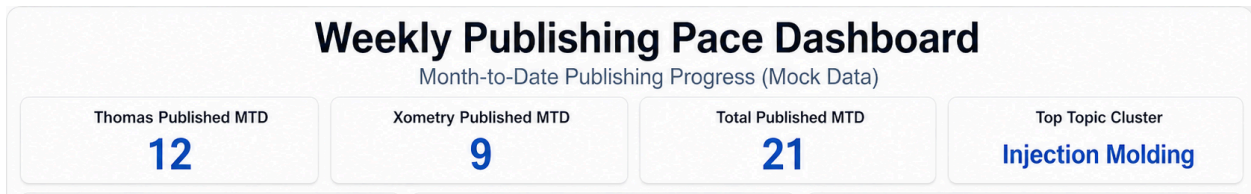
A flowchart showing the decision process:

URL present → status is Published → publish date is in current month → include in weekly month-to-date dashboard.



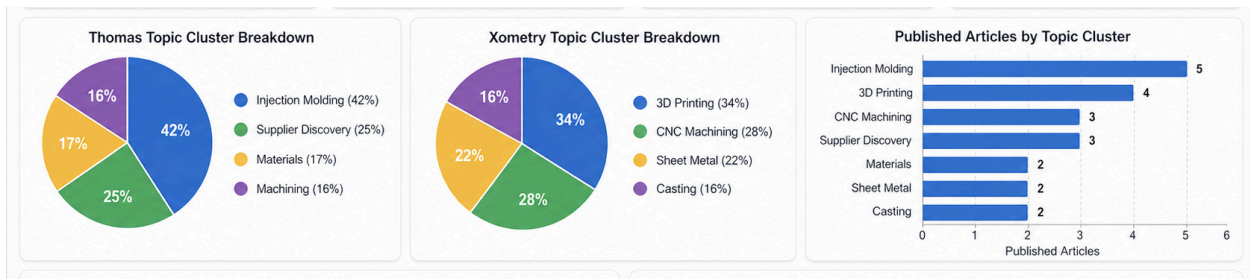
Visual 4: Mock Brand Scorecards

A mockup showing each brand’s month-to-date publishing totals, such as total published articles, number of topic clusters, and top topic cluster.



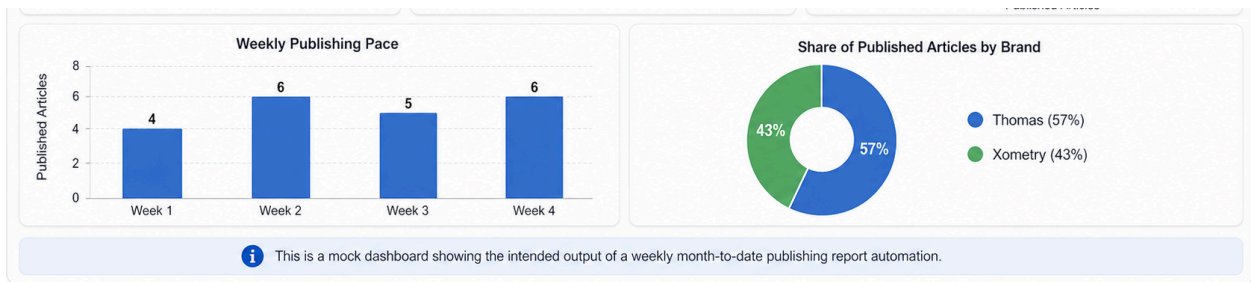
Visual 5: Mock Dashboard Visualization

A mockup showing pie charts for topic cluster breakdown by brand and/or a bar chart of total articles by topic cluster.



Visual 6: Weekly Publishing Pace Chart

A bar chart or line chart showing how many articles were published each week of the current month, and a pie chart showing the share of published articles by brand.



5. Automation Strategy & Python Toolkit

| Library Category | Library Name | Functional Purpose |
|------------------|--------------|--------------------|
|                  |              |                    |

|                           |                                                     |                                                                                                                              |
|---------------------------|-----------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------|
| Data Manipulation         | <code>pandas</code>                                 | Load Google Sheets data into DataFrames, filter rows, combine multiple trackers, and group by brand, topic cluster, and week |
| Date Logic                | <code>datetime</code>                               | Identify the current month and apply month-to-date filtering automatically                                                   |
| Google Sheets Integration | <code>gsread</code>                                 | Connect to source Google Sheets, pull tracker data, and write the dashboard output back to Google Sheets                     |
| Google Authentication     | <code>google.colab.auth</code> / Google credentials | Authenticate access to Google Sheets when running the script in Google Colab                                                 |
| Visualization             | <code>matplotlib</code> / <code>plotly</code>       | Create scorecards, pie charts, bar charts, and publishing pace charts                                                        |
| File/Report Output        | Google Sheets tabs                                  | Store the cleaned dataset, summary tables, and dashboard outputs in organized Google Sheet tabs                              |

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## 6. Logic Preview

```

from datetime import datetime

current_month = datetime.now().strftime("%Y-%m")

if article_url != "" and status == "Published" and
publish_date.startswith(current_month):
    print("Include in weekly month-to-date dashboard.")
else:
    print("Exclude or flag for review.")

```

This preview shows the basic rule the future automation would follow: only rows with a valid URL, a “Published” status, and a publish date in the current month should be included in the weekly month-to-date dashboard.

A more complete version of the automation would loop through each row in the source Google Sheets, apply these rules, group the qualifying rows by brand, topic cluster, and publish week, and then create dashboard visuals.

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## 7. Dashboard Output Preview

The final automated output would be a dashboard-style report in Google Sheets rather than only a manually filtered table. It would summarize publishing activity and show where the month stands as of the weekly report date.

| Dashboard Element        | Business Question Answered                                    |
|--------------------------|---------------------------------------------------------------|
| Brand scorecards         | How much has each brand published so far this month?          |
| Article share by brand   | What percentage of month-to-date output came from each brand? |
| Topic cluster pie charts | What topics made up each brand’s publishing mix?              |
| Topic cluster bar chart  | Which topics had the most publishing activity overall?        |
| Weekly publishing trend  | Is publishing pace consistent or concentrated in one week?    |

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## 8. Input / Output

### Inputs

The inputs are Google Sheets-based article tracking records with fields such as:

- `article_url`
- `topic_cluster`
- `publish_date`
- `status`
- Google Sheet name or source tab

### Outputs

The outputs are Google Sheet tabs containing:

- a cleaned month-to-date dataset
- brand scorecards
- topic cluster summaries
- pie chart data tables
- weekly publishing pace summaries
- dashboard visuals

The process is complete when the weekly dashboard shows current month-to-date publishing performance by brand, topic cluster, and publishing pace.